An Introduction to Linear Algebra

Linear
- lines
- we can visualize in $\mathbb{R}^2$, $\mathbb{R}^3 \sim \mathbb{R}^n$?

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Examples. A line in $\mathbb{R}^2$ is given by

$$ax + by = c.$$

A plane in $\mathbb{R}^3$ is given by

$$ax + by + cz = d.$$

A linear equation in n variables is of the form

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = b.$$

★ Addition & Scalar Multiplication

→ $x_1, x_2, \ldots, x_n$: variables

→ $a_1, a_2, \ldots, a_n \in \mathbb{R}$: coefficients

Ex.
- $x + 3y - 7 = 0$ is linear. $x + 3y = 7$
- $x_1 - 2x_2 - 3x_3 + x_4 = 0$ is linear.
- $x + 3y^2 = 4$ NOT linear.

We can construct a system of linear equations:

- $\begin{cases} 5x + y = 3 \\ 2x - y = 4 \end{cases}$ 2 variables x, y; 2 equations
- $\begin{cases} 4x_1 - x_2 + 3x_3 = -1 \\ 3x_1 + x_2 - 9x_3 = -4 \end{cases}$ 3 variables; 2 equations
- More generally:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

m equations, n unknowns (variables)

Goal: Solve linear systems!
We find values of $x_1, \ldots, x_n$ that satisfy all the equations in the system.

If a linear system has at least one solution, then the system is consistent. Otherwise, the system is inconsistent. If there is exactly one solution, then the solution is said to be unique.

Q. How do we solve linear systems?

Ex. Let's solve $\begin{cases} 5x + y = 3 & (1) \\ 2x - y = 4 & (2) \end{cases}$.

* 2 equations, 2 unknowns

→ Substitution: (1) $y = 3 - 5x$

⇝ (2) $2x - (3 - 5x) = 4$

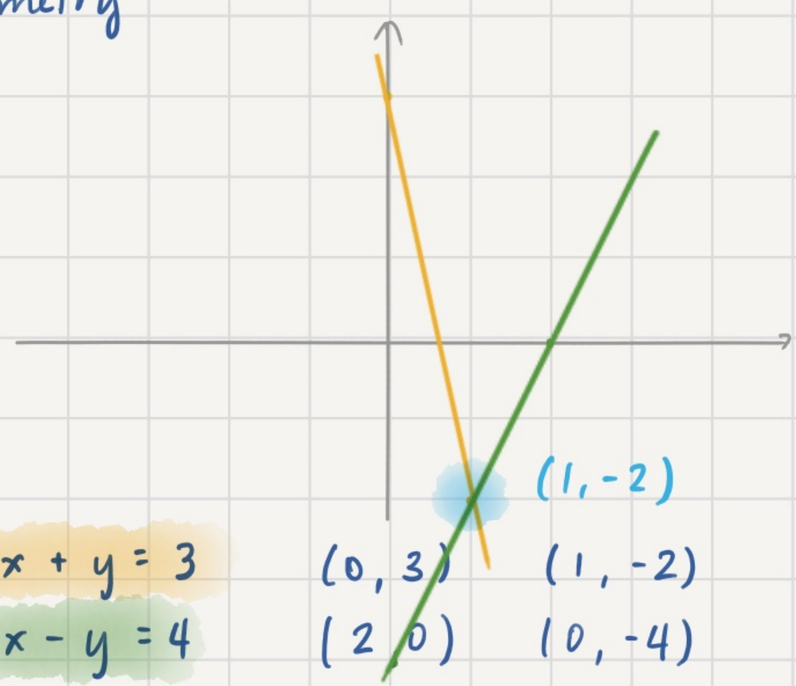
$\Rightarrow 2x - 3 + 5x = 4$

$\Rightarrow 7x = 7$

So, $x = 1$ and $y = 3 - 5(1) = -2$.

* Only useful for small systems!

→ Geometry



$5x + y = 3$	$(0, 3)$	$(1, -2)$
$2x - y = 4$	$(2, 0)$	$(0, -4)$

The solution is $x = 1$, $y = -2$.

* Only useful in 2 (maybe 3) variables!

→ Elimination

$$\begin{cases} 5x + y = 3 \\ 2x - y = 4 \end{cases} \xrightarrow{E_2 + E_1} \begin{matrix} 5x + y = 3 \\ 7x + 0 = 7 \end{matrix}$$

$$\xrightarrow{\frac{1}{7}E_2} \begin{cases} 5x + y = 3 \\ x = 1 \end{cases}$$

$$\xrightarrow{E_1 - 5E_2} \begin{cases} 0 + y = -2 \\ x = 1 \end{cases} \Rightarrow \begin{cases} x = 1 \\ y = -2 \end{cases}$$

* Systematic! We can generalize this to larger systems!

Rules of Elimination

- We can scale an equation by a non-zero constant
- We can add a multiple of one equation to another.
- We can interchange a pair of equations

Idea: Work on $\boxed{\text{matrices}}$ instead!

$$\begin{array}{c} \begin{array}{cc} \text{"}x\text{"} & \text{"}y\text{"} \end{array} \\ \left(\begin{array}{cc|c} 5 & 1 & 3 \\ 2 & -1 & 4 \end{array}\right) \end{array} \rightarrow \cdots \rightarrow \left(\begin{array}{cc|c} 0 & 1 & -2 \\ 1 & 0 & 1 \end{array}\right)$$

$\rightsquigarrow x = 1, y = -2$

Operations are performed on rows.

Consider a linear system:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \qquad\qquad \vdots \qquad\qquad \vdots \qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}.$$

Construct an augmented matrix:

$$\begin{array}{c} \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right) \\ \begin{array}{ccccc} x_1 & x_2 & \cdots & x_n & \text{"="} \end{array} \end{array}.$$

Elementary Row Operations

① Scale a row by a non-zero constant:

$$cR_i, \quad c \neq 0$$

② Add a scalar multiple of one row to another row:

$$R_i + cR_j, \quad i \neq j$$

③ Swap any two rows:

$$R_i \leftrightarrow R_j$$

Ex. Let's solve

$$\begin{cases} x + y + 2z = 9 \\ 2x + 4y - 3z = 1 \\ 3x + 6y - 5z = 0 \end{cases} \rightarrow \begin{cases} x = 1 \\ y = 2 \\ z = 3 \end{cases}$$

Construct the augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right)$$

☆ Try to eliminate the variables!

$$\xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right)$$

$$\xrightarrow{\frac{1}{2}R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{array}\right)$$

$$\xrightarrow{R_3 - 3R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{array}\right)$$

$$\xrightarrow{-2R_3} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right) \rightarrow z = 3$$

② $y - \frac{7}{2}z = -\frac{17}{2} \Rightarrow y = -\frac{17}{2} + \frac{7}{2}(3) = 2$

① $x + y + 2z = 9 \Rightarrow x = 9 - (2) - 2(3) = 1$

Ex. Consider $\begin{cases} x + y = 4 \\ 3x + 3y = 6 \end{cases}$.

$$\left(\begin{array}{cc|c} 1 & 1 & 4 \\ 3 & 3 & 6 \end{array}\right) \xrightarrow{R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & 0 & -6 \end{array}\right)$$

$0 = -6$

This is a contradiction!

This is an inconsistent linear system: it has no solution!

Ex. Consider $\begin{cases} 4x - 2y = 1 \\ 16x - 8y = 4 \end{cases}$.

$$\left(\begin{array}{cc|c} 4 & -2 & 1 \\ 16 & -8 & 4 \end{array}\right) \xrightarrow{R_2 - 4R_1} \left(\begin{array}{cc|c} 4 & -2 & 1 \\ 0 & 0 & 0 \end{array}\right)$$

$0 = 0$

So, $4x - 2y = 1$. Let $y = t$:

$$x = \frac{1 + 2t}{4}, \quad y = t, \quad t \in \mathbb{R}.$$

This has infinitely many solutions!

Theorem 1. Every linear system will have either no solution, a unique solution, or infinitely many solutions. There are no other possibilities.